

EDITOR	Ramchandra Chandra
EDITORIAL CORRESPONDENCE	Editorial Secretary Phone: 011-24818096 Email: info@efyindia.com Technical queries: efy@efyindia.com
SUBSCRIPTIONS & MISSING ISSUES	Customer Care Officer, EFY Phone: 011-24818061 Email: niranjan@efyindia.com
BACK ISSUES, BOOKS, CDs, PCRs, etc.	80's Space D-80/5, Okhla Industrial Area, Phase-1 New Delhi 110020 Phone: 011-24818061, 24817461, 24817462 Email: kishu@efyindia.com
EXCLUSIVE NEWS/STORY DISTRIBUTOR	India Book House Pvt Ltd Ach. No. 35, Behar Mohan Bridge Mahabares, Mumbai 400014 Phone: 022-23541463, 23541481, 40497411, 40497412, Fax: 24952092 Email: info@bhworld.com
ADVERTISEMENTS NEW DELHI (HEAD OFFICE)	Hargal Bhat D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020, INDIA Phone: 011-24818062, 24818063 Fax: 24817543 Email: anil@efyindia.com
MUMBAI	Fairy D'Sa 207, Shreevati Chambers 40/3, A.R. Road, West Mumbai Mumbai 400018, Ph: 022-24950047, 24928250 Fax: 24954278 Email: efy@efyindia.com
BANGALORE	Jayashree No. 9, 7th Main, 1st Cross, HAL II Stage Indiranagar, Bangalore 560038 Ph: 080-2246504, 2246505, Fax: 22465094 Email: efy@efyindia.com
CHENNAI	M. Nandanan JES Road, 37-A, Cathedral Garden Road Near Palmgrove Hotel, Chennai 600034 Ph: 044-26275191, Mobile: 9896250248 Email: efy@efyindia.com
KOLKATA	D.C. Mukherjee 61-A, Park Street, Kolkata 700016 Ph: 033-2294308, Mobile: 22620094 Email: efy@efyindia.com Mobile: 9943242952
PUNE	Zakir Shabbir Email: efy@efyindia.com Mobile: 9957240753
HYDERABAD	P.S. Muradharan Mobile: 9849436460
GUJARAT	Shambhushank Patel Ph: 079-22138441 Email: info@efyindia.com Mobile: 9825213841
SINGAPORE	M. Pooja's Day Publications Singapore Pte Ltd 72, Bendemeer Road, #02-20, The Laurens, Singapore 339941 Phone: +65-6836 2272 Fax: +65-6287 7302 Email: info@efyindia.com, singapore@efyindia.com
UNITED STATES	McVernigan Lannan E & Tech Media Phone: +1 800 536 6407 Email: verniganlannan@gmail.com

Printed, published and owned by Ramchandra Chandra. Printed at Nandani Press, 30, Okhla Industrial Area, Phase-1, New Delhi on the first day of each month and published from D-87/1, Okhla Industrial Area, Phase-1, New Delhi 110020. Copyright 2009. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, the responsibility of any error is solely for any loss to the publisher. Articles that are not used are returned to the author. Accompanying a self-addressed and sufficiently stamped envelope. But no responsibility is taken for any loss or damage returning the material. EFY will not be responsible for any wrong claims made by an advertisement. If any will be within a New Delhi country only.

Period	SUBSCRIPTION RATES*			
	Domestic	Digital	Overseas	
Year	With CD	Without CD	(With CD)	(Without CD)
Five	Rs 2,160	Rs 1,445	—	—
Three	1,510	1,015	—	—
One	575	385	US\$ 100	US\$ 65

For receiving Annual Directories (print) at Rs 300 with CD, please add Rs 250 for each year (including handling and forwarding).
Monthly add Rs 50/- for outside India delivery.
Please send payments only in favour of EFY Enterprises Pvt Ltd.
Non-empt of copies may be reported to support@efyindia.com mentioning your subscription number.

*For anniversary offer, refer to page Nos. 116 & 117

CONTENTS

28	EFY Report Electronics Industry Forecasts for Year 2009
36	Industry Vision India In Year 2008, Future Ahead!
44	Survey When the Going Gets Tough The Tough Get Going
52	Technology Focus E-waste Management In India
60	Careers Get Embedded Right Away!
64	Education Learning on Mobile Anytime, Anywhere
68	Construction Designing Low-Power Products Using MSP430 Microcontroller (Part 1 of 2)
76	Construction Sequential Power-On*
99	Circuit Ideas Top 10 Circuits
110	Software Section Desktop Publishing Utility Program in Visual Basic
115	Technology Focus Plasma Arc Gasification For Waste Management (Part 1 of 2)
124	Security Using Tech Eyes To Spot Terrorists
128	Satcom Early Warning System for Landslides
140	China Electronics Fair Opportunities Galore For Those Who Can Grab Them
164	Interview "OLEDs represent the future of display technology"— Satish Kaura, CMD, Samtel
166	Interview "Automobiles offer a huge opportunity"— Kai Wang, VP of sales & marketing and GM of the Asia-Pacific Region for Freescale

Departments

EFY DVD	08
LETTERS	16
Qs&As	17
TECHNOLOGY NEWS	20
CIRCUIT IDEAS	94
NEW PRODUCTS	143
INDUSTRY NEWS	146
BIZ ADS	151
ELECTRONICS MART	156
BOOKSHELF	167
SUBJECT & ORGANISATION INDEX	168
ADVERTISERS' INDEX	169

eStyle

- 80 First Look!**
- High-speed Gaming Workstations
 - HD Camcorder with 7.6cm LCD
 - Affordable Multimedia Phone
 - Stylish CDMA Phones
 - Multimedia Notebook
 - Affordable Notebook
 - Rugged Mobile
 - Entry-Level A4 Scanner
 - 3G Version of HTC Touch

- 86 Buyers' Guide**
Which Camera Phone To Buy?

NEXT ISSUE

Technology Focus:
Artificial Space Satellites

Buyers' Guide:
Which DTH?

Survey/EFY Report:
Budget Memorenda